

Sr.No	Module	Weightage %	Marks Weightage%
1	Module-V	20.00	21.05
2	Module-I	20.00	19.74
3	Module-II	20.00	19.74
4	Module-III	20.00	19.74
5	Module-IV	20.00	19.74
Weightage representation okay : Weightage representation exceeded more than 5% :Weightage representation exceeded more than 10%			

CSE601

[ET]

Enroll. No. - _____

END SEMESTER EXAMINATION : April-May, 2026

Data Structures and Algorithm Design

Time : 3.00 Hrs

Maximum Marks : 60

Note: Attempt questions from all sections as directed.

Section - A : Attempt any Four questions out of Five . Each question carries 06 marks.

Q1.

(a)

Define Asymptotic Notations: Big-O, Big-mega, and Big-Theta.

(b)

Explain the applications of a Double Linked List.

(3)

Delete

Q2.

(a)

Differentiate between Depth First Search (DFS) and Breadth First Search (BFS).

(b)

What are the properties of a Red-Black Tree?

(3)

Delete

Q3.

(a)

Differentiate between BST and AVL Tree.

(b)

Discuss about Nolish Notation.

(3)

Delete

Q4.

Compare and contrast the Dynamic Programming approach with the Greedy Approach.

(6)

Delete

Q5.

(a)

Differentiate between NP-Hard and NP-Complete classes.

(b)

What is a SAT problem?

(3)

Delete

Section – B : Attempt any two questions out of three. Each question carries 10 marks.

Q6.

(a)

Explain the Divide and Conquer paradigm.

(b)

Describe the efficiency of the Quick Sort algorithm.

(5)

Delete

Q7.

(a)

Illustrate the working of Prim’s Algorithm for finding the Minimum Spanning Tree.

(b)

Explain the "All-pair Shortest Path" problem and mention the algorithm used to solve it.

(5)

Delete

Q8.

(a)

Solve the 0/1 Knapsack problem using Dynamic Programming for a given set of weights and values.

(b)

Explain the rotations required in AVL Tree with example.

(5)

Delete

Section - C : Compulsory question

Q9.

(a)

Discuss the efficiency of different sorting techniques: Bubble, Insertion, and Heap Sort.

(b)

Compare 0/1 Knapsack (DP) with Fractional Knapsack.

(c)

Solve the Longest Common Subsequence (LCS) for two given strings: X = ABCBDAB, Y = BDCABA.

(d)

Explain the Greedy choice property.

(4)

Delete

(4)

Delete

(4)

Delete

(4)

Delete